

EBN122 CLASS TEST / KLASTOETS 2 (23 SEP 2016)**ANSWER SHEET / ANTWOORD BLAD**

Surname:	MEMO	Student number:							
Initials:		AFR (Eng 3-6)	ENG (Eng 3-1)	ENG (Eng 3-6)					

Total:	/15
---------------	------------

1	$V_{0(30V)} = -60V$	
2	$V_0 = 4V$	
3	$V_{th} = 2,67V$	$R_{th} = -0,67\Omega$
4	$R_N = 10\Omega$	
5	$V_0 = 8,5V$	
6	$V_1 = 5,5V$	
7	$I_S = -4mA$	

DO NOT remove this page from the test.

EBN122 Class Test / Klastoets 2 (23 September 2016)

15 Marks/Punte

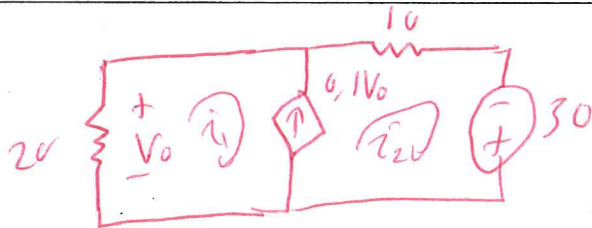
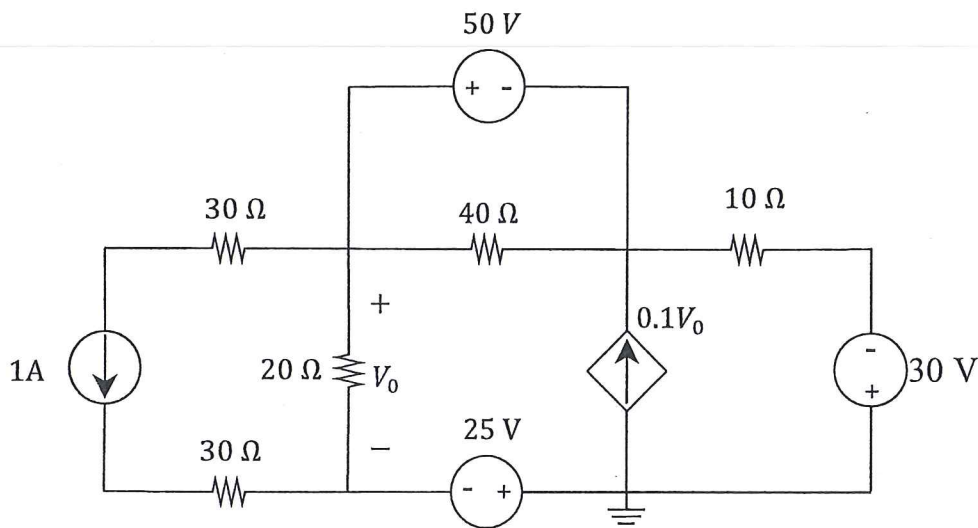
30 Minutes/Minute

Question 1 [2]

Using superposition, find the individual contribution of the 30V voltage source to V_0 in the circuit below.

Vraag 1 [2]

Gebruik superposisie, en bepaal die individuele bydra van die 30 V spanningsbron tot V_0 .



$$V_0 = -i_1(20)$$

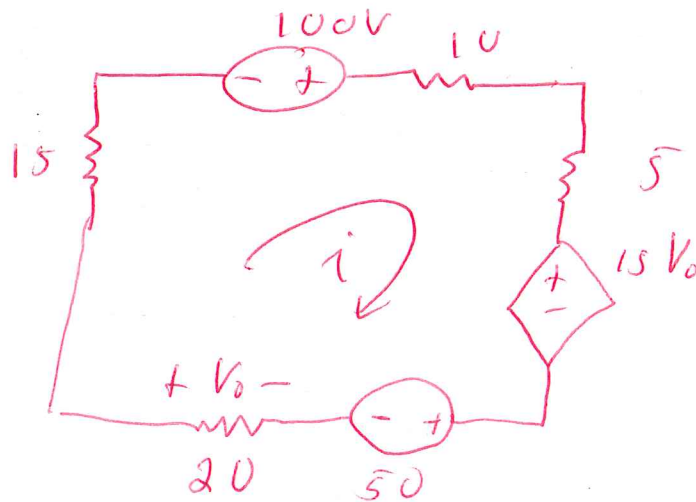
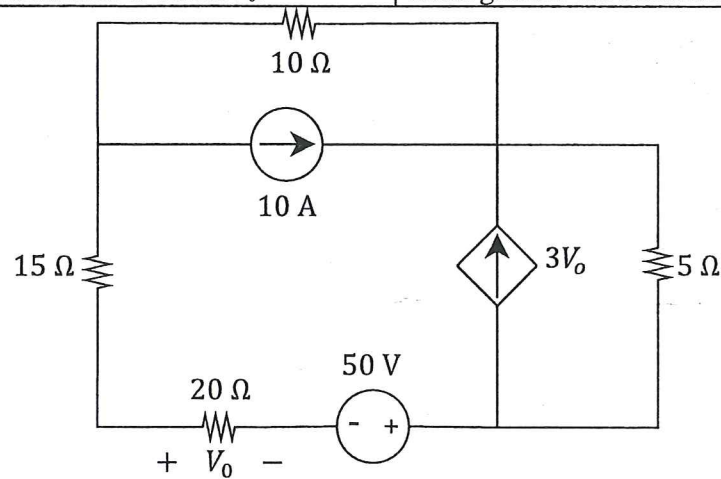
$$\begin{aligned} \cancel{i_1(20)} \quad 0.1V_0 + i_1 &= i_2 \\ (0.1)(-i_1)(20) + i_1 &= i_2 \\ \therefore -i_1 &= i_2 \end{aligned}$$

$$20i_1 + 10i_2 - 30 = 0$$

$$20i_1 - 10i_1 = 30$$

$$\therefore i_1 = 3A$$

$$\therefore V_0 = -3(20) = -60V$$

Question 2 [2]Make use of source transformation to obtain V_o .**Vraag 2 [2]**Maak gebruik van brontransformatie om V_o te bepaal.

$$V_o = -20i$$

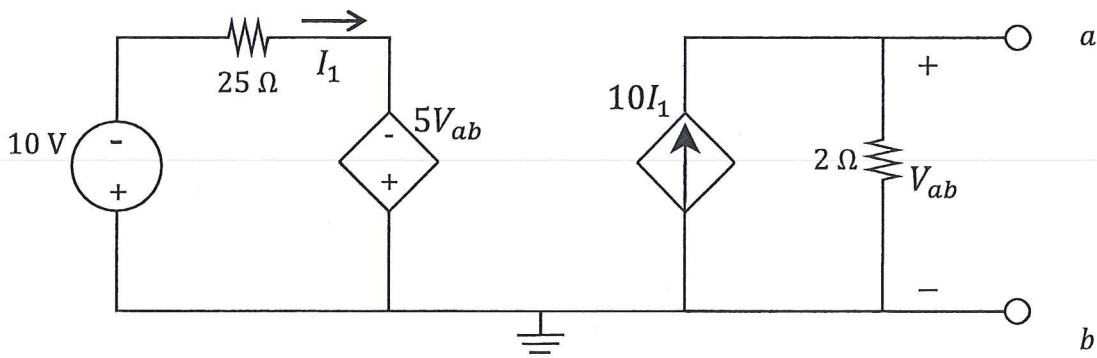
$$50i - 50 + 15V_o = 0$$

$$50i - 50 - 300i = 0$$

$$-250i = 50$$

$$i = -0,2$$

$$\therefore V_o = -20(-0,2) = 4V$$

Question 3 [4]Find the Thevenin equivalent with respect to terminals $a-b$.**Vraag 3 [4]**Bepaal die Thevenin ekwivalent met verwysing tot terminale $a-b$.

$$I_T = \frac{1}{2} - 10I_1$$

$$= \frac{1}{2} - 10 \left(\frac{5}{25} \right)$$

$$= -\frac{3}{2}$$

$$R_{TH} = -\frac{2}{3} = -0.667 \Omega$$

$$V_{ab} = 10 I_1 (2) = 20 I_1$$

$$10 + 25 I_1 - 5 V_{ab} = 0$$

$$25 I_1 - 100 I_1 = -10$$

$$\therefore I_1 = \frac{2}{15}$$

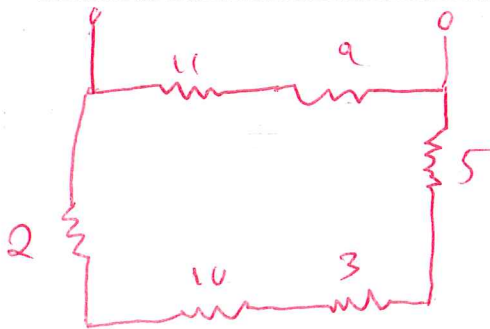
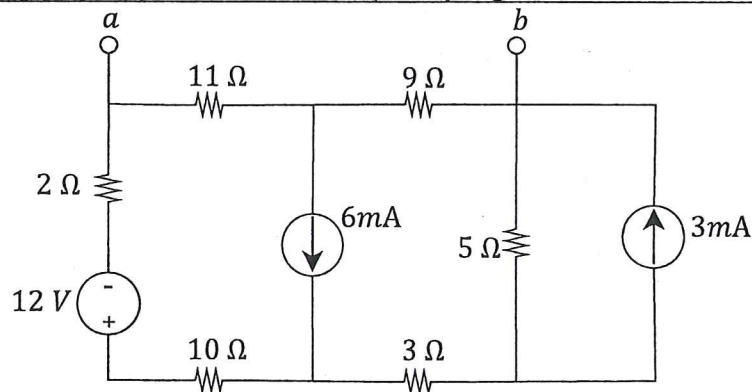
$$V_{ab} = 20 \left(\frac{2}{15} \right) = 2.67 V$$

Question 4 [2]

Determine the Norton equivalent resistance R_N with respect to terminals $a-b$.

Vraag 4 [2]

Bepaal die Norton ekwivalente weerstand R_N met verwysing tot terminale $a-b$.



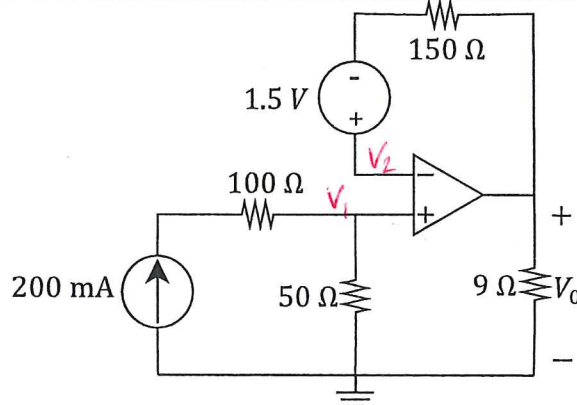
$$R_N = (2 + 10 + 3 + 5) \parallel (11 + 9) = 10 \Omega$$

Question 5 [2]

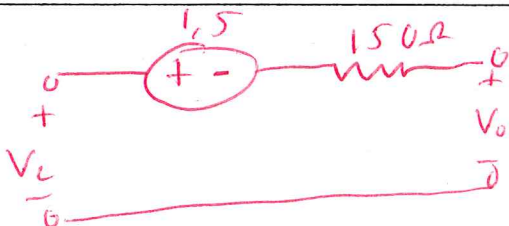
Obtain V_o for the op amp circuit below.

Vraag 5 [2]

Bepaal V_o vir onderstaande kringbaan.



$$V_1 = V_2 = 50(200 \times 10^{-3}) = 10V$$

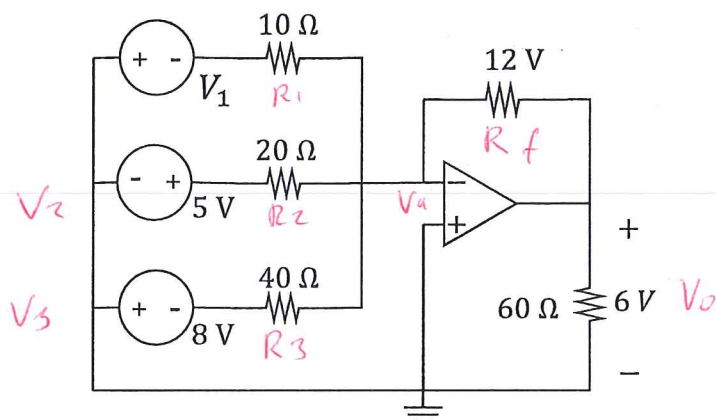


$$\begin{aligned} -V_1 + 1.5 + 150(0) + V_o &= 0 \\ -10 + 1.5 + V_o &= 0 \\ \therefore V_o &= 8.5V \end{aligned}$$

Question 6 [2]

 Obtain V_1 for the op amp circuit below.

Vraag 6 [2]

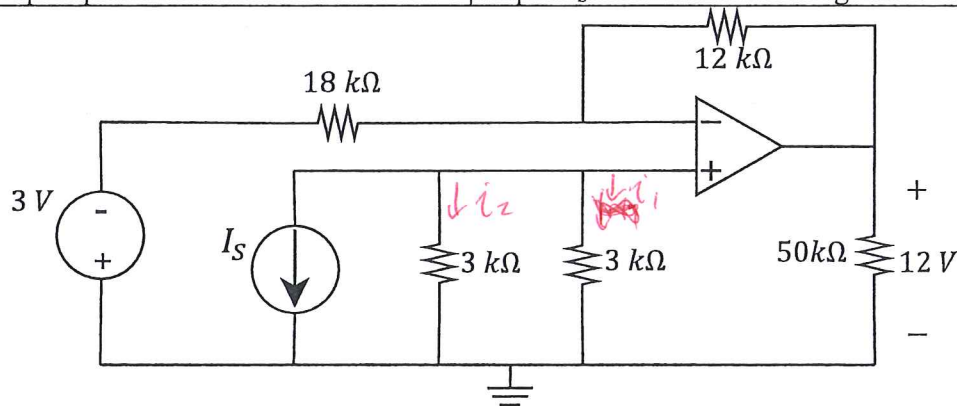
 Bepaal V_1 vir onderstaande kringbaan.


$$\begin{aligned} \frac{V_a - V_o}{R_f} &= \frac{-V_1 - V_a}{R_1} + \frac{V_2 - V_a}{R_2} + \frac{-V_3 - V_a}{R_3} \\ -\frac{V_o}{R_f} &= -\frac{V_1}{R_1} + \frac{V_2}{R_2} - \frac{V_3}{R_3} \\ -\frac{6}{12} &= -\frac{V_1}{10} + \frac{5}{20} - \frac{8}{40} \\ -0,55 &= -\frac{V_1}{10} \\ \therefore V_1 &= 5,5 \text{ V} \end{aligned}$$

Question 7 [2]

 Obtain I_S for the op amp circuit below.

Vraag 7 [2]

 Bepaal I_S vir onderstaande kringbaan.


$$\begin{aligned} \frac{V_a - 12}{12k} &= \frac{-3 - V_o}{18k} \\ 18V_a - 216 &= -36 - 12V_a \\ 30V_a &= 180 \\ V_a &= 6 \end{aligned}$$

$$\begin{aligned} i_1 &= i_2 = \frac{6}{3k} = 0,002 \\ I_S + 0,002 + 0,002 &= 0 \\ \therefore I_S &= -4 \text{ mA} \end{aligned}$$